[0x22]

let $y := 0x\,aaaaaaaaaaaaaaab$

① unsigned multiply : $rdx = rdx \cdot rax \rightarrow y \cdot rax \in rdx : rax$

upper bits

lower bits

②

$y \cdot rax_0 \in (rdx >> 1) : rax$

$\overset{③}{\Vert}$

rax

$$f(x) = \text{floor}\left(\frac{x}{3}\right)$$

```
① { mov     rdx, 0xaaaaaaaaaaaaaaab
  { mul     rdx
② shr       rdx, 1
  mov       rax, rdx
```

$rdx = rdx_0 \cdot rcx_0$ $\in$ 64 bits unsigned

$rdx /= 2 \Rightarrow rdx = \frac{rdx_0 \cdot rcx}{2}$

$rax = rdx$
$= \frac{rdx_0 \cdot rcx}{2}$

~~ie $rax = rdx_0 \cdot rcx$~~

~~$= \frac{(0x\,aaaaaaaa\,aaaaaaab) \cdot rcx}{2}$~~

~~ie $(rdx \cdot rcx) >> 1$~~

~~ie $(0x\,aaaaaaaa\,aaaa\,aaab \cdot rcx) >> 1$~~

just run $\in$ gdb (empirically :))

f(0x40)	0x15	64 → 21	
f(0x9)	0x3	10 → 3	
f(0x30)	0x10	ie 48 → 16	⇒ f(x) = floor(x/3)
f(0x20)	0x9	32 → 10	
f(0x9)	0x3	9 → 3	
f(0x8)	0x2 [0x22]	8 → 2	